

Ex. No. : 02

Date : 31/07/2026

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Data Modelling and Form Design

Aim:

To design and implement data models and forms in Zoho Creator for the student management application, by creating appropriate forms with required fields, field types, and validations, and to establish the relationships between them for efficient data storage and retrieval.

Procedure:

1. Sign in to Zoho Creator
 - Open a web browser and go to creator.zoho.com.
 - Log in using a valid Zoho account (or sign up if you don't have one).
2. Create a New Application
 - Click on **Create New Application**.
 - Choose **Create from scratch** (or use a template if available).
 - Enter the **Application Name** and a short description.
 - Click **Create**.
3. Data Modelling — Plan the Structure
 - Identify the entities required for the application (e.g., Student, Course, Employee, Department, etc.).
 - Determine the attributes (fields) for each entity along with their data types (Text, Number, Date, Email, Decimal, Dropdown, Lookup, etc.).

Identify relationships between entities:

- **One-to-Many** (e.g., one Department has many Employees) — implemented using a **Lookup field**.
 - **Many-to-Many** — implemented using a **Multi Select Lookup** field or a separate linking form.
4. Create Forms
 - Inside the application, click **Add Form**.
 - Give the form a suitable **Form Name** (e.g., "Student Details").
 - Drag and drop the required fields from the field palette onto the form:
 - Single Line/Multi Line for text
 - Number/Decimal for numeric values
 - Date/Date-Time for dates
 - Dropdown/Radio/Checkbox for selection fields

- Set properties for each field: Label name, Mandatory/Optional, Unique value, Default value, Validation rules (e.g., regex, character limit).
 - Rearrange fields using drag-and-drop for a proper layout.
 - Click **Save** to save the form.
5. Establish Relationships
- Add a **Lookup field** in the child form pointing to the parent form to create a one-to-many relationship.
 - Verify the relationship by checking that related records appear correctly linked.
6. Preview and Test the Form
- Click on **Access Form / Preview** to view the form as an end user.
 - Enter sample data and submit to test field validations and relationships.
7. Save and Deploy
- Save all changes made to the form and application.
 - Publish/deploy the application so that users can start entering data.

Screenshots:

The screenshot displays a web application interface for a 'Student Management System'. The main content area is titled 'Academic Record' and contains a form with the following fields:

- Student Name ***: Two input fields for 'First Name' and 'Last Name'.
- Program Name**: A single input field.
- Enrollment Date**: An input field with a date mask 'dd-MMM-yyyy'.
- Academic Level**: Three radio button options: 'First Year', 'Second Year', and 'Third Year'.
- GPA**: An input field with a mask '#####'.
- Semester**: An input field with a mask '#####'.
- Course Code**: An input field with a mask '#####'.

The left sidebar contains a navigation menu with the following items:

- Student Management...
- Academic Record (selected)
- Academic Record Report
- Guardian Contact
- Course Enrollment
- Student Profile

The top bar of the application shows the title 'Student Management System' and includes a trial status 'Trial expires in 8 days' with an 'Upgrade' link, and buttons for 'Edit this application' and 'Help'.

← → ↻ creatorapp.zoho.in/harininagarajan.2024.cse1/student-management-system#Form:Guardian_Contact ☆ School ⋮

Student Management System Trial expires in 8 days Upgrade Edit this application Help

Student Management...

- Academic Record >
- Guardian Contact
- Guardian Contact**
- Guardian Contact Report
- Course Enrollment >
- Student Profile >

Guardian Contact

Student Name
First Name Last Name

Guardian Name
First Name Last Name

Relationship

Contact Number

Email

Address

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← → ↻ creatorapp.zoho.in/harininagarajan.2024.cse1/student-management-system#Form:Course_Enrollment ☆ School ⋮

Student Management System Trial expires in 8 days Upgrade Edit this application Help

Student Management...

- Academic Record >
- Guardian Contact >
- Course Enrollment**
- Course Enrollment Report
- Student Profile >

Course Enrollment

Student *
First Name Last Name

Course Code *

Course Name *

Semester

Enrollment Date

Instructor Name

Credits

Grade

The screenshot displays a web browser window with the URL `creatorapp.zoho.in/harininagarajan.2024.cse1/student-management-system#Form:Student_Profile`. The application is titled "Student Management System" and has a trial expiration notice: "Trial expires in 8 days". The left sidebar contains the following navigation items:

- Academic Record
- Guardian Contact
- Course Enrollment
- Student Profile (selected)
- Student Profile Report

The main content area shows the "Student Profile" form with the following fields:

- First Name:
- Last Name:
- Date of Birth:
- Gender:
- Contact Number:
- Email:
- Enrollment Status:
- Academic Year:
- Program of Study:

Result:

Thus, the data model and forms for the given application were successfully designed and implemented in Zoho Creator, with appropriate fields, validations, and relationships established between forms.